

Sliding Window Filtering for Robotic Navigation

Mitchell Cohen

Department of Mechanical Engineering, McGill University
817 Sherbrooke Street West, Montreal QC H3A 0C3

November 24, 2025

Contents

1	Sliding Window Filtering	1
1.1	Batch MAP Estimation	2
1.2	Marginalization of Old States	4

1 Sliding Window Filtering

The purpose of this document is to present the sliding window filter, a commonly used online, constant-time estimation algorithm that is an approximation to the full batch maximum a posteriori (MAP) problem. The sliding window filter was first introduced in [1], and has been utilized as the backbone in many estimation pipelines, for example, visual-inertial odometry, [2], [3] and lidar-inertial odometry [4]. Sliding window filters offer an optimization-based alternative to filtering algorithms such as the extended Kalman filter (EKF), which linearize the process and measurement models only a single time. Sliding window filters perform batch estimation over a sliding window of states, and are hence able to relinearize process and measurement models, offering the benefit of reducing the linearization errors that the EKF suffers from.

These notes are primarily based on [5], which derived a sliding window filter for the basic 2D SLAM problem with odometry and landmark measurements. To start, the full batch MAP estimator will be presented, which is at the core of the sliding window filtering algorithm. Then, the *marginalization* step of the sliding window filter will be derived, allowing the approximation of the full MAP estimator over a window of states.

1.1 Batch MAP Estimation

Consider a state trajectory of a system given by

$$\mathcal{X} = (\mathcal{X}_0, \dots, \mathcal{X}_K), \quad (1)$$

where $\mathcal{X}_k \in G$ is the state of the system at time $t = t_k$. The system state is often, for example, the pose of a robot, along with some additional parameters for estimation (i.e., sensor biases, velocities, etc.), and belongs to an arbitrary Lie group G . Throughout this document, similar notation to [6] will be used for handling states on Lie groups.

The goal of MAP estimation is to estimate the entire trajectory of the system given the following three pieces of information:

1. A *prior distribution* on the initial state, $\hat{\mathcal{X}}_0$, which is assumed Gaussian such that $\mathcal{X}_0 \sim \mathcal{N}(\hat{\mathcal{X}}, \mathbf{P}_0)$, where $\mathbf{P}_0$ is the initial covariance,
2. The *process model* $\mathcal{X}_k = f(\mathcal{X}_{k-1}, \mathbf{u}_{k-1}, \mathbf{w}_{k-1})$, where $\mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{k-1})$ is Gaussian noise, and $\mathbf{u}_{k-1}$ is a measured noisy input, such as odometry measurements. This input can be used to predict the system state forward from time $t = t_{k-1}$ to $t = t_k$.
3. The *measurement model* $\mathbf{y}_k = \mathbf{g}(\mathcal{X}_k, \mathbf{v}_k)$, where $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$ is Gaussian measurement noise.

Given the entire sequence of inputs and measurements from time $t = t_0$ to $t = t_K$, denoted $\mathcal{U} = \{\mathbf{u}_0, \dots, \mathbf{u}_{K-1}\}$, and $\mathcal{Y} = \{\mathbf{y}_0, \dots, \mathbf{y}_K\}$ respectively, the MAP estimate is a point estimate that solves

$$\hat{\mathcal{X}}^{\text{MAP}} = \arg \max_{\mathcal{X}} p(\mathcal{X} | \hat{\mathcal{X}}_0, \mathcal{U}, \mathcal{Y}) \quad (2)$$

$$= \arg \max_{\mathcal{X}} \alpha p(\mathcal{Y} | \mathcal{X}, \hat{\mathcal{X}}_0, \mathcal{U}) p(\mathcal{X} | \mathcal{X}_0, \mathcal{U}), \quad (3)$$

where Bayes rule has been used to rewrite the original problem, and the denominator has been lumped into a constant α as the denominator of Bayes rule does not depend on $\mathcal{X}$. Under the assumptions of Gaussian measurement likelihoods, maximizing the posterior is equivalent to minimizing the nonlinear least squares cost function given by

$$J(\mathcal{X}) = J^{\text{prior}} + \sum_{i=1}^k J_i^u + \sum_{i=0}^k J_i^y, \quad (4)$$

$$J^{\text{prior}}(\mathcal{X}_0) = \frac{1}{2} \|\mathbf{e}_0(\mathcal{X}_0)\|^2, \quad (5)$$

$$J_i^u(\mathcal{X}_{i-1}, \mathcal{X}_i) = \frac{1}{2} \|\mathbf{e}_i^u(\mathcal{X}_{i-1}, \mathcal{X}_i)\|_{\mathbf{Q}_i}^2 \quad (6)$$

$$J_i^y(\mathcal{X}_i) = \frac{1}{2} \|\mathbf{e}_i^y(\mathcal{X}_i)\|_{\mathbf{R}_k}^2, \quad (7)$$

where the notation $\|\mathbf{e}\|_{\mathbf{M}}^2 = \mathbf{e}^\top \mathbf{M}^{-1} \mathbf{e}$, and the prior, process and measurement model error terms are given by

$$\mathbf{e}_0(\mathcal{X}_0) = \mathcal{X}_0 \ominus \hat{\mathcal{X}}_0, \quad (8)$$

$$\mathbf{e}_i^u(\mathcal{X}_{i-1}, \mathcal{X}_i) = \mathcal{X}_i \ominus f(\mathcal{X}_{i-1}, \mathbf{u}_{i-1}, \mathbf{w}_{i-1}), \quad (9)$$

$$\mathbf{e}_i^y(\mathcal{X}_i) = \mathbf{y}_i - \mathbf{g}(\mathcal{X}_i, \mathbf{v}_i), \quad (10)$$

and $\ominus : G \times G \rightarrow \mathbb{R}^n$ is a difference between two Lie group elements as defined in [6]. For each error term, the squared Mahalanobis distance $\|\mathbf{e}\|_{\mathbf{M}}^2 = \mathbf{e}^\top \mathbf{M}^{-1} \mathbf{e}$ can be equivalently written as

$$\tilde{\mathbf{e}}^\top \tilde{\mathbf{e}} = \mathbf{e}^\top \mathbf{L} \mathbf{L}^\top \mathbf{e}, \quad (11)$$

where $\mathbf{M} = \mathbf{L} \mathbf{L}^\top$ is the Cholesky decomposition of $\mathbf{M}$. By multiplying each error term by the square root of the inverse covariance as $\tilde{\mathbf{e}} = \mathbf{L}^\top \mathbf{e}$, the correlations between their errors are removed and the variance of the error becomes identity. This is a process known as *whitening* the error. By stacking all whitened error terms into a single error vector given by

$$\tilde{\mathbf{e}} = \begin{bmatrix} \tilde{\mathbf{e}}_0^\top & \tilde{\mathbf{e}}_1^{u\top} & \cdots & \tilde{\mathbf{e}}_k^{u\top} & \tilde{\mathbf{e}}_k^{y\top} & \cdots & \tilde{\mathbf{e}}_k^{y\top} \end{bmatrix}^\top \quad (12)$$

the original MAP problem can be transformed into an unweighted nonlinear least squares problem as

$$\hat{\mathcal{X}} = \arg \min_{\mathcal{X}} = \frac{1}{2} \tilde{\mathbf{e}}(\mathcal{X})^\top \tilde{\mathbf{e}}(\mathcal{X}). \quad (13)$$

Nonlinear least squares problems in the form of (13), rely on a *linearization* of the whitened error $\tilde{\mathbf{e}}(\mathcal{X})$ with respect to $\mathcal{X}$, where the Jacobian of the error is written as

$$\mathbf{H} = \left. \frac{D\tilde{\mathbf{e}}(\mathcal{X})}{D\mathcal{X}} \right|_{\mathcal{X}=\bar{\mathcal{X}}}. \quad (14)$$

Iterative nonlinear least squares methods such as Gauss-Newton or Levenberg-Marquardt, utilize this error Jacobian to compute an update to the state variables. For example, Gauss-Newton computes a step as

$$\delta \boldsymbol{\xi} = (\mathbf{H}^\top \mathbf{H})^{-1} \mathbf{H}^\top \tilde{\mathbf{e}}, \quad (15)$$

which is then used to update the state estimate as

$$\hat{\mathcal{X}} \leftarrow \hat{\mathcal{X}} \oplus \delta \boldsymbol{\xi}. \quad (16)$$

The process of linearizing the error about the current estimate of the state, computing the step (15), and updating the state using (16) repeats until convergence to yield the MAP estimate $\hat{\mathcal{X}}$.

1.2 Marginalization of Old States

As the robot moves through the environment, recording new inputs and collecting new measurements the size of the state vector $\mathcal{X}$ would continue growing indefinitely, quickly making the full batch MAP estimate infeasible to compute in real-time. To design a constant-time algorithm suitable for providing real-time state estimates, the sliding window filter employs *marginalization* of past states. To detail the process of marginalization, consider the following scenario: the robot travels, collecting measurements from times $t \in [t_0, t_k]$. Then, the oldest m states are removed from the problem via marginalization, such that the set of states to marginalize out is given by $\mathcal{X}_m = (\mathcal{X}_0, \dots, \mathcal{X}_{m-1})$. Consider next the set of states connected to $\mathcal{X}_m$ via error terms, denoted $\mathcal{X}_r$. This set of states is sometimes referred to as the *Markov blanket* in literature. Finally, the set of states not involved in error terms involving $\mathcal{X}_m$ is denoted $\mathcal{X}_n$. At time $t = t_k$, the MAP cost function can be written as

$$J(\mathcal{X}) = J_{mr}(\mathcal{X}_m, \mathcal{X}_r) + J_{rn}(\mathcal{X}_r, \mathcal{X}_n), \quad (17)$$

where the cost term J_{mr} contains all terms involving states $\mathcal{X}_m$ and $\mathcal{X}_r$ only, and J_{rn} contains all terms involving states $\mathcal{X}_r$ and $\mathcal{X}_n$ only. Since states marginalized at time $t = t_k$ do not participate in any measurements after that time, there do not exist terms jointly involving states in $\mathcal{X}_n$ and $\mathcal{X}_m$. The decomposition of the cost function can then be utilized to reformulate the minimization problem as

$$\min_{\mathcal{X}_m, \mathcal{X}_r, \mathcal{X}_n} J(\mathcal{X}) = \min_{\mathcal{X}_r, \mathcal{X}_n} \left(\min_{\mathcal{X}_m} J(\mathcal{X}_m, \mathcal{X}_r, \mathcal{X}_n) \right), \quad (18)$$

$$= \min_{\mathcal{X}_r, \mathcal{X}_n} \left(J_{rn}(\mathcal{X}_r, \mathcal{X}_n) + \min_{\mathcal{X}_m} J_{mr}(\mathcal{X}_r, \mathcal{X}_m) \right), \quad (19)$$

where the first equality holds for any multivariate optimization problem. Starting with the minimization of J_{mr} with respect to $\mathcal{X}_m$, at time $t = t_k$, consider the quadratic approximation of $J_{mr}(\mathcal{X}_m, \mathcal{X}_r)$ about the current estimates $\hat{\mathcal{X}}_m(t_k)$ and $\hat{\mathcal{X}}_r(t_k)$, written as

$$J_{mr}(\mathcal{X}_m, \mathcal{X}_r) \approx J_{mr}(\hat{\mathcal{X}}_m(t_k), \hat{\mathcal{X}}_r(t_k)) + \mathbf{b}^\top(t_k) \delta \boldsymbol{\xi}(t_k) + \frac{1}{2} \delta \boldsymbol{\xi}^{k^\top} \mathbf{A}(t_k) \delta \boldsymbol{\xi}(t_k) \quad (20)$$

$$\mathbf{b}(t_k) = \begin{bmatrix} \mathbf{b}_m(t_k) \\ \mathbf{b}_r(t_k) \end{bmatrix}, \quad \mathbf{A}(t_k) = \begin{bmatrix} \mathbf{A}_{mm}(t_k) & \mathbf{A}_{mr}(t_k) \\ \mathbf{A}_{rm}(t_k) & \mathbf{A}_{rr}(t_k) \end{bmatrix}, \quad \delta \boldsymbol{\xi}(t_k) = \begin{bmatrix} \delta \boldsymbol{\xi}_m(t_k) \\ \delta \boldsymbol{\xi}_r(t_k) \end{bmatrix}, \quad (21)$$

where the $\mathcal{X}$ is parameterized as $\mathcal{X} = \hat{\mathcal{X}} \oplus \delta \boldsymbol{\xi}$, the states estimates for $\mathcal{X}_m$ and $\mathcal{X}_r$ at time $t = t_k$ are denoted $\hat{\mathcal{X}}_m(t_k)$ and $\hat{\mathcal{X}}_r(t_k)$ respectively, and $\mathbf{b}(t_k)$ and $\mathbf{A}(t_k)$ are the gradient and the Hessian of the cost function, evaluated at $\hat{\mathcal{X}}_m$ and $\hat{\mathcal{X}}_r$, respectively. Additionally, the error $\delta \boldsymbol{\xi}(t_k)$ is given by

$$\delta \boldsymbol{\xi}(t_k) = \mathcal{X}(t_k) \ominus \hat{\mathcal{X}}(t_k), \quad (22)$$

where $\ominus : G \times G \rightarrow \mathbb{R}^n$ is a difference between two Lie group elements. The next step involves eliminating $\delta \boldsymbol{\xi}_m$ from the quadratic expansion of the cost function in (23). The perturbation $\delta \boldsymbol{\xi}$ that minimizes the cost can be found in closed form by taking the gradient of the quadratic cost with

respect to $\delta\boldsymbol{\xi}$ and setting equal to zero. The cost (23) can be expanded as

$$J(\mathcal{X}_m, \mathcal{X}_r) \approx J(\hat{\mathcal{X}}_m, \hat{\mathcal{X}}_r) + \mathbf{b}_m^\top \delta\boldsymbol{\xi}_m + \mathbf{b}_r^\top \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_m^\top \mathbf{A}_{mm} \delta\boldsymbol{\xi}_m + \delta\boldsymbol{\xi}_m^\top \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{rr} \delta\boldsymbol{\xi}_r. \quad (23)$$

The gradient of the quadratic expansion with respect to $\delta\boldsymbol{\xi}_m$ is given by

$$\frac{\partial J}{\partial \delta\boldsymbol{\xi}_m} = \mathbf{b}_m + \mathbf{A}_{mm} \delta\boldsymbol{\xi}_m + \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r. \quad (24)$$

Setting equal to zero yields the a perturbation $\delta\boldsymbol{\xi}_m$ that minimizes the quadratic expansion, written as

$$\delta\boldsymbol{\xi}_m^* (\delta\boldsymbol{\xi}_r) = -\mathbf{A}_{mm}^{-1} (\mathbf{b}_m + \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r). \quad (25)$$

To yield an expression for the cost that is independent of $\mathcal{X}_m$, substitute this back into the cost in (23), and expand to yield a cost *independent* of $\delta\boldsymbol{\xi}_m$. Subbing into (23) term by term yields

$$\mathbf{b}_m^\top \delta\boldsymbol{\xi}_m = -\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{b}_m - \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r, \quad (26)$$

$$\begin{aligned} \frac{1}{2} \delta\boldsymbol{\xi}_m^\top \mathbf{A}_{mm} \delta\boldsymbol{\xi}_m &= \frac{1}{2} (\mathbf{b}_m + \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r)^\top \mathbf{A}_{mm}^{-1} (\mathbf{b}_m + \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r), \\ &= \frac{1}{2} \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{b}_m + \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{b}_m + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r, \end{aligned} \quad (27)$$

$$\delta\boldsymbol{\xi}_m^\top \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r = (\mathbf{b}_m + \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r)^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r, \quad (28)$$

$$= -\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r - \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r \quad (29)$$

Next, all terms can be combined and grouped by order. The constant term is given by

$$\alpha = J(\hat{\mathcal{X}}_m, \hat{\mathcal{X}}_r) - \frac{1}{2} \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{b}_m. \quad (30)$$

There are three terms that are linear in $\delta\boldsymbol{\xi}$, given by $\mathbf{L}_1 = -\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r$, $\mathbf{L}_2 = \frac{1}{2} \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{b}_m$, and $\mathbf{L}_3 = -\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r$. The linear terms sum to

$$\mathbf{L} = \mathbf{L}_1 + \mathbf{L}_2 + \mathbf{L}_3, \quad (31)$$

$$= -\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r + \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r - \mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r \quad (32)$$

$$= -\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r. \quad (33)$$

Next, the terms that are quadratic in $\delta\boldsymbol{\xi}_r$ are given by

$$\mathbf{Q} = -\frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r. \quad (34)$$

Combining all terms, including the ones that were in in the original cost function yields

$$J \approx \alpha + (-\mathbf{b}_m^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r + \mathbf{b}_r^\top \delta\boldsymbol{\xi}_r) + \left(-\frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr} \delta\boldsymbol{\xi}_r + \delta\boldsymbol{\xi}_r^\top \mathbf{A}_{rr} \delta\boldsymbol{\xi}_r \right), \quad (35)$$

$$= \alpha + (\mathbf{b}_r - \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{b}_m)^\top \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top (\mathbf{A}_{rr} - \mathbf{A}_{mr}^\top \mathbf{A}_{mm}^{-1} \mathbf{A}_{mr}) \delta\boldsymbol{\xi}_r, \quad (36)$$

$$= \alpha + \mathbf{b}_p^\top \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_p \delta\boldsymbol{\xi}_r. \quad (37)$$

where α is a constant term, $\delta\boldsymbol{\xi}_r = \mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k)$ is the error between the remaining state and the estimate at time $t = t_k$, and $\mathbf{A}_p(t_k)$ and $\mathbf{b}_p(t_k)$ are given by

$$\mathbf{A}_p(t_k) = \mathbf{A}_{rr}(t_k) - \mathbf{A}_{rm}(t_k)\mathbf{A}_{mm}(t_k)^{-1}\mathbf{A}_{mr}(t_k) \quad (39)$$

$$\mathbf{b}_p(t_k) = \mathbf{b}_r(t_k) - \mathbf{A}_{rm}(t_k)\mathbf{A}_{mm}(t_k)^{-1}\mathbf{b}_m(t_k) \quad (40)$$

Note that the matrices $\mathbf{A}_p(t_k)$ and $\mathbf{b}_p(t_k)$ are evaluated at the state estimate at time $t = t_k$. Combining this with the original minimization in (18), the minimization of the cost function $J(\mathcal{X}_m, \mathcal{X}_r, \mathcal{X}_n)$ with respect to the entire history of states is approximately equivalent to the minimization of the cost function $J'(\mathcal{X}_r, \mathcal{X}_n)$, with respect to $\mathcal{X}_r$ and $\mathcal{X}_n$, written as

$$J'(\mathcal{X}_r, \mathcal{X}_n) = J_{\text{marg}}(\mathcal{X}_r) + \frac{1}{2} \sum_{i \in \mathcal{S}_{k'}} \mathbf{e}_i(\mathcal{X}_r, \mathcal{X}_n)^\top \mathbf{e}_i(\mathcal{X}_r, \mathcal{X}_n), \quad (41)$$

$$J_{\text{marg}}(\mathcal{X}_r) = \mathbf{b}_p(t_k)^\top \left(\mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right) + \frac{1}{2} \left(\mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right)^\top \mathbf{A}_p(t_k) \left(\mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right), \quad (42)$$

where $\mathcal{S}_{k'}$ is the set of measurements involved with states $\mathcal{X}_r$ and $\mathcal{X}_n$ only. Note that now the cost function is *independent* of the marginalized state $\mathcal{X}_m$. The cost $J_{mr}(\mathcal{X}_m, \mathcal{X}_r)$ has been approximated by its second-order Taylor expansion, and the information has been absorbed into a new cost $J_{\text{marg}}(\mathcal{X}_r)$, which is only a function of $\mathcal{X}_r$ as well as the state estimate of $\mathcal{X}_r$ at time $t = t_k$. This approximation induces small errors to the MAP problem, but is able to bound the state size. The minimization of the new cost function can be performed then be performed with respect to $(\mathcal{X}_r, \mathcal{X}_m)$ on the new window.

Note that the form of the marginalization cost in (42) cannot be directly used within nonlinear least squares solvers. The marginalization cost can also be written as

$$J_{\text{marg}}(\mathcal{X}_r) = \mathbf{b}_p(t_k)^\top \left(\mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right) + \frac{1}{2} \left\| \mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right\|_{\mathbf{A}_p(t_k)}^2, \quad (43)$$

which has both a quadratic and linear term. In order to use this cost term within existing nonlinear least squares frameworks, it must be rewritten as quadratic function. Consider the residual definition given by

$$J_{\text{marg}}(\mathcal{X}_r) = \frac{1}{2} \left\| \mathbf{A}_m \left(\mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right) + \mathbf{b}_m \right\|^2, \quad (44)$$

$$\mathbf{A}_m^\top \mathbf{A}_m = \mathbf{A}_p(t_k), \quad (45)$$

$$\mathbf{A}_m^\top \mathbf{b}_m = \mathbf{b}_p(t_k). \quad (46)$$

Expanding the residual yields

$$\frac{1}{2} (\mathbf{A}_m \delta\boldsymbol{\xi}_r + \mathbf{b}_m)^\top \left(\mathbf{A}_m \delta\boldsymbol{\xi}_r + \frac{1}{2} \mathbf{b}_m \right) = \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_m^\top \mathbf{A}_m \delta\boldsymbol{\xi}_r + \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_m^\top \mathbf{b}_m + \frac{1}{2} \mathbf{b}_m^\top \mathbf{A}_m \delta\boldsymbol{\xi}_r + \frac{1}{2} \mathbf{b}_m^\top \mathbf{b}_m, \quad (47)$$

$$= \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_m^\top \mathbf{A}_m \delta\boldsymbol{\xi}_r + \mathbf{b}_m^\top \mathbf{A}_m \delta\boldsymbol{\xi}_r + \frac{1}{2} \mathbf{b}_m^\top \mathbf{b}_m, \quad (48)$$

$$= \frac{1}{2} \delta\boldsymbol{\xi}_r^\top \mathbf{A}_p \delta\boldsymbol{\xi}_r + \mathbf{b}_p^\top \delta\boldsymbol{\xi}_r + \text{const}, \quad (49)$$

showcasing that the residual in (44) yields the same cost as (42), up to an additive constant. Therefore, after marginalization of parameters $\mathcal{X}_m$, when the next window is optimized, the following residual and Jacobian should be added to the problem.

$$\mathbf{e}_{\text{marg}}(\mathcal{X}_r) = \mathbf{A}_m \left(\mathcal{X}_r \ominus \hat{\mathcal{X}}_r(t_k) \right) + \mathbf{b}_m, \quad (50)$$

$$\mathbf{H}_{\text{marg}} = \mathbf{A}_m. \quad (51)$$

References

- [1] G. Sibley, L. Matthies, and G. Sukhatme, “Sliding Window Filter with Application to Planetary Landing,” *Journal of field robotics*, vol. 27, no. 5, pp. 587–608, 2010.
- [2] T. Qin, P. Li, and S. Shen, “VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator,” *IEEE transactions on robotics*, vol. 34, no. 4, pp. 1004–1020, 2018.
- [3] C. Chen, P. Geneva, Y. Peng, W. Lee, and G. Huang, “Optimization-based VINS: Consistency, Marginalization, and FEJ,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023, pp. 1517–1524.
- [4] T. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, and D. Rus, “LIO-SAM: Tightly-Coupled Lidar-Inertial Odometry via Smoothing and Mapping,” in *2020 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, IEEE, 2020, pp. 5135–5142.
- [5] T.-C. Dong-Si and A. I. Mourikis, “Motion Tracking with Fixed-Lag Smoothing: Algorithm and Consistency Analysis,” in *2011 IEEE International Conference on Robotics and Automation*, IEEE, 2011, pp. 5655–5662.
- [6] J. Sola, J. Deray, and D. Atchuthan, “A Micro Lie Theory for State Estimation in Robotics,” *arXiv preprint arXiv:1812.01537*, 2018.